

Chapter 1

Sources and Generation of Surface Contaminants and Their Impact

Rajiv Kohli

The Aerospace Corporation, NASA Johnson Space Center, Houston, Texas, USA

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1 INTRODUCTION

Contaminants are ubiquitous in nature as by-products of natural processes as well as by-products of human and industrial activities. A contaminant is any material, substance, or energy that is unwanted or adversely affects the product [1]. Surface contamination, the subject of this chapter, can be in many forms and may be present in a variety of states on the surface. For example, typical surface defects on hard disk drive (HDD) components include roughness and wear, flaking, porosity, loose debris, nonvolatile residue (NVR), flash, bloom and inclusions on plastics, burrs and rust on metals, lubricant leakage in ball bearings, packaging debris, and uncured adhesives [2,3]. A particle with dimensions similar to the flying height between the recording head and the disk that is trapped between the head and the disk in a disk drive can cause catastrophic failure of the drive [3–6]. Ionic contaminants also can lead to corrosion of the recording head and failure of the drive [7]. In semiconductor wafer

processing, airborne molecular contaminants (AMCs) can cause various defects, such as haze on the wafer surface or introduction of foreign dopants, which can lead to loss of the entire product yield [8–10]. The most common categories of surface contaminants are given below.

- Particles such as dust, metals, ceramics, glass, and plastics in the submicrometer to macro size range
- Thin film or molecular contamination that can be organic (hydrocarbons) or inorganic (acids gases, bases)
- Cationic (such as Na^+ and K^+) and anionic (such as Cl^- , F^- , Br^- , SO_3^- , BO_3^{3-} , and PO_4^{3-}) contamination
- Microbial contamination (bacteria, fungi, and algae).

Other contaminant categories include toxic and hazardous chemicals, radioactive materials, and by-products of manufacturing processes, which are identified on surfaces employed in specific industries, such as metals processing, chemical production, nuclear industry, pharmaceutical manufacture, and food processing, handling, and delivery. Energy sources or changes of state, such as light, electric and magnetic fields, heat, and radiation, can also be contaminating. For example, light is usually desirable, but in the photographic film development process, it becomes a contaminant. Magnetic fields can contaminate iron and nickel magnets. Many plastics, when exposed to ultraviolet light, slowly release inert particles due to their degradation. Common contamination sources include machining oils and greases, hydraulic and cleaning fluids, adhesives, waxes, human contamination, and particulates. In addition, a whole host of other chemical contaminants from a variety of sources may also soil a surface. The emphasis in this chapter will be on particles as contaminants deposited on the surface of products manufactured in controlled environments such as clean rooms. Aerosol particles and their behavior will not be considered here. There are numerous references, including books and journals, which address all aspects of aerosols, including single aerosol particles [11–24]. The discussion in this chapter is necessarily an overview of the subject matter and no attempt is made at comprehensiveness. Each of the thousands of references to contamination, cleanliness, and cleaning published each year mentions some aspect of the sources and generation of contaminants, making it nearly impossible to comprehensively treat the subject matter in this chapter. The references cited do provide sources of additional information.

2 SURFACE CLEANLINESS LEVELS

Surface cleaning specifications are typically based on the amount of specific or characteristic contaminant remaining on the surface after it has been cleaned. Space agencies worldwide specify surface precision cleanliness levels for space hardware by particle size (in the micrometer (μm) size range) and number of particles, as well by film contamination represented by NVR [25,26]. The cleanliness levels are based on contamination levels established in industry standard IEST-STD-CC1246D (which replaced the original cleanliness

standard MIL-STD-1246 [27]) for particles from Level 1 to Level 1000 and for NVR from Level AA5 (10 ng/0.1 m²) to Level J (25 mg/0.1 m²) as shown in Table 1.1 [28]. The cleanliness levels have been revised or redesignated in revision E of this standard [29]. The maximum allowable number of particles for each particle size range has been rounded in revision E, while the NVR

TABLE 1.1 Product Surface Cleanliness Levels for Commercial and Noncommercial Applications Defined in IEST-STD-CC1246 Revision D [28]				
Cleanliness Level	Particulate Level		NVR Level	
	Particle Size (µm)	Maximum Allowable Count Per 0.1 m ² /or 0.1 L of Gas or Liquid	Level	Quantity Mass/0.1 m ² or Mass/0.1 L (Gas or Liquid)
1	1	1	AA5	10 ng
5	1	2.8	AA4.7	20 ng
	2	2.3	AA4.3	50 ng
	5	1	AA4	100 ng
10	1	8.4	AA3.7	200 ng
	2	6.9	AA3.3	500 ng
	5	2.9	AA3	1 µg
	10	1	AA2.7	2 µg
25	2	53.1	AA2.3	5 µg
	5	22.7	A/100	10 µg
	15	13.3	A/50	20 µg
	25	1	A/20	50 µg
50	5	166	A/10	100 µg
	15	24.6	A/5	200 µg
	25	7.2	A/2	500 µg
	50	1	A	1 mg
100	5	1780	B	2 mg
	15	264	C	3 mg
	25	78.4	D	4 mg
	50	10.7	E	5 mg
	100	1	F	7 mg

Continued

TABLE 1.1 Product Surface Cleanliness Levels for Commercial and Noncommercial Applications Defined in IEST-STD-CC1246 Revision D [28]—Cont'd

Cleanliness Level	Particulate Level		NVR Level	
	Particle Size (μm)	Maximum Allowable Count Per 0.1 m^2 /or 0.1 L of Gas or Liquid	Level	Quantity Mass/ 0.1 m^2 or Mass/ 0.1 L (Gas or Liquid)
200	15	4180	G	10 mg
	25	1230	H	15 mg
	50	169	J	25 mg
	100	15.8		
	200	1		
300	25	7450		
	50	1020		
	100	95		
	250	2.2		
	300	1		
500	50	11,800		
	100	1090		
	250	26.3		
	500	1		
750	50	95,800		
	100	8910		
	250	213		
	500	8.1		
	750	1		
1000	100	42,600		
	250	1020		
	500	38.7		
	750	4.7		
	1000	1		

designation levels have been replaced with a single letter R followed by the maximum allowable mass of NVR. For example, former NVR level J has the new designation R25; level A/2 is now R5E-1; and level AA5 is now R1E-5. However, this revision has not yet been officially adopted by the space agencies. [Table 1.2](#) lists the surface cleanliness levels for particle contamination in revision E of IEST-STD-CC1246. [Table 1.3](#) compares NVR contamination designation levels between revisions D and E of the standard.

Revision E of IEST-STD-CC1246 provides two additional categories for specification of surface cleanliness levels.

- Particle fallout specification is important in many space programs for sensitive hardware exposed to clean room environments for extended periods. [Table 1.4](#) lists particle fallout counts for different cleanliness levels. The specification method uses an “F” descriptor to designate the fallout requirement. As an example, a specification “IEST-STD-CC1246 Level 100+F100” indicates an initial surface cleanliness Level 100 verified per [Table 1.2](#) and Level 100 as the additional particle contamination allowed due to fallout per [Table 1.4](#). In this case, the final cleanliness level of the hardware must meet the particle contamination level shown in column 3 of [Table 1.5](#).
- Visibly clean (VC) is frequently specified for surface cleanliness of spaceflight hardware free of all visible contamination, including particles and NVR. This is a qualitative specification. Several different VC cleanliness levels can be specified based on incident light illumination intensity and the viewing distance from the surface. [Table 1.6](#) defines the VC levels in IEST-STD-CC1246E, which also shows the equivalence with NASA Johnson Space Center (JSC) standard [26,30] developed for the manned spaceflight program.

The full description of surface cleanliness level is specified as IEST-STD-CC1246E Level X Y Z. Here X is the numerical particulate cleanliness level from [Table 1.2](#); Y is the NVR cleanliness level from [Table 1.3](#); and Z is an alternative or additional cleanliness level, such as particle fallout from [Table 1.5](#) or VC cleanliness level from [Table 1.6](#). Some examples are shown in [Table 1.7](#).

In several industries, the precision cleanliness level is defined as an organic contaminant level of less than 10 μg of contaminant per cm^2 , although many applications are setting the requirement at less than 1 $\mu\text{g}/\text{cm}^2$ [28,29]. These cleanliness levels are either very desirable or are required by the function of parts such as machined parts, electronic assemblies, optical and laser components, precision mechanical parts, and computer parts.

Many of the products and manufacturing processes are also sensitive to, or they can even be destroyed by, AMC that are present due to external, process, or otherwise generated sources, making it essential to monitor and control these contaminants. AMC is chemical contamination in the form of vapors or aerosols that can be organic or inorganic, and it includes everything from acids and bases to organometallic compounds and dopants [31,32]. A new standard ISO 14644-10, “Clean rooms and associated controlled environments—Part 10: Classification

TABLE 1.2 Product Surface Particle Cleanliness Levels for Commercial and Noncommercial Applications Defined in IEST-STD-CC1246 Revision E [29]

Size Bins		Maximum Particle Count Per 0.1 m ² for Each Cleanliness Level								
Min (μm)	Max (μm)	25	50	100	200	300	400	500	750	1000
5	15	19	141	1519						
15	25	2	17	186	2949					
25	50	1	6	67	1069	6433				
50	100	0	1	9	154	926	3583	20726		
100	250	0	0	1	15*	92	359	1073	9704	
250	500	0	0	0	0	2*	8*	25	205	983
500	750	0	0	0	0	0	0	1	7	33
750	1000	0	0	0	0	0	0	0	1	3
1000	1250	0	0	0	0	0	0	0	0	1

*Under no circumstances is more than one particle larger than the level designation allowed per 0.1 m² of surface area.

TABLE 1.3 Comparison of Surface NVR Cleanliness Level Designations for Commercial and Noncommercial Applications Defined in IEST-STD-CC1246 Revisions D and E [28,29]

Designation Revision D	Designation Revision E	Maximum Allowable NVR Limit Mass/0.1 m ² or Mass/0.1 L (Gas or Liquid)
AA5	R1E-5	10 ng
AA4.7	R2E-5	20 ng
AA4.3	R5E-5	50 ng
AA4	R1E-4	100 ng
AA3.7	R2E-4	200 ng
AA3.3	R5E-4	500 ng
AA3	R1E-3	1 µg
AA2.7	R2E-3	2 µg
AA2.3	R5E-3	5 µg
A/100	R1E-2	10 µg
A/50	R2E-2	20 µg
A/20	R5E-2	50 µg
A/10	R1E-1	100 µg
A/5	R2E-1	200 µg
A/2	R5E-1	500 µg
A	R1	1 mg
B	R2	2 mg
C	R3	3 mg
D	R4	4 mg
E	R5	5 mg
F	R7	7 mg
G	R10	10 mg
H	R15	15 mg
J	R25	25 mg

TABLE 1.4 Particle Fallout Cleanliness Levels for Commercial and Noncommercial Applications Defined in IEST-STD-CC1246 Revision E [29]

Size Bins		Maximum Particle Count Per 0.1 m ² for Each Fallout Cleanliness Level								
<i>Min (μm)</i>	<i>Max (μm)</i>	25	50	100	200	300	400	500	750	1000
5	15	2	5	14	47					
15	25	1	2	5	15	33	58			
25	50	1	1	4	12	27	49	78		
50	100		1	2	6	13	23	37	91	
100	250			1	3	6	10	16	41	80
250	500					1	2	3	8	15
500	750							1	1	3
750	1000								1	1
1000	1250									1

TABLE 1.5 Total Particle Fallout Cleanliness Levels for Commercial and Noncommercial Applications Defined in IEST-STD-CC1246 Revision E [29]

Size Bins		Maximum Particle Count Per 0.1 m ² for Fallout Cleanliness Level		
<i>Min (μm)</i>	<i>Max (μm)</i>	<i>100</i>	<i>F100</i>	<i>100 + F100</i>
5	15	1519	14	1533
15	25	186	5	191
25	50	67	4	71
50	100	9	2	11
100	250	1	1	2

of surface cleanliness by chemical concentration” [33] is now available as an international standard that defines the classification system for cleanliness of surfaces in clean rooms with respect to the presence of chemical compounds or elements (including molecules, ions, atoms, and particles). The standard identifies at least eight (8) categories of AMCs—acid (ac), base (ba), biotoxic (bt), condensable (cd), corrosive (cr), dopant (dp), total organic compounds (toc), and oxidant (ox), as well as individual substances or groups of substances. This standard is

TABLE 1.6 Criteria for Different VC Cleanliness Levels in IEST-STD-CC1246E and the Equivalence to the NASA JSC Standard

IEST-STD-CC1246E			NASA JSC Standard		
<i>Cleanliness Level</i>	<i>Viewing Distance*</i>	<i>Light Intensity*</i>	<i>Cleanliness Level</i>	<i>Viewing Distance*</i>	<i>Light Intensity*</i>
VC-0.5-1000	0.25-0.5 m (~9.84-19.68 in.)	1000 lumens/m ² (~93 ft-candles)	Visibly Clean Highly Sensitive (VC-HS)	6-18 in. (~0.15-0.46 m)	100 ft-candles (~1076 lumens/m ²)
VC-1-500	0.5-1.0 m (~1.64-3.28 ft.)	500 lumens/m ² (~46 ft-candles)	Visibly Clean Sensitive (VC-S)*	2-4 ft. (~0.61-1.22 m)	50 ft-candles (~538 lumens/m ²)
VC-2-500	1.0-2.0 m (~3.28-6.56 ft.)	500 lumens/m ² (~46 ft-candles)	No equivalence	–	–
VC-3-500	1.5-3.0 m (~4.92-9.84 ft.)	500 lumens/m ² (~46 ft-candles)	Visibly Clean Standard (VC)*	5-10 ft. (~1.52-3.04 m)	50 ft-candles (~538 lumens/m ²)

*The viewing distance and light intensity (white light) in the NASA JSC standard are given in US customary units (feet and inches; foot-candles) and are given in metric units in IEST-STD-CC1246E. The converted values are listed in parentheses below the original units for each standard.

TABLE 1.7 Descriptors for Cleanliness Level for Particles, NVR, and An Alternate Cleanliness Parameter as Defined in Revision E of Standard IEST-STD-CC1246			
X	Y	Z	Full Description
300			IEST-STD-CC1246E Level 300 Specifies: particles only
100 + F100			IEST-STD-CC1246E Level 100+F100 Specifies: particles and accumulated fallout
200	R2		IEST-STD-CC1246E Level 200 R2 Specifies: particles and NVR
		VC-0.5- 1000UV	IEST-STD-CC1246E Level VC-0.5-1000UV Specifies: visibly clean using UV light
		VC-3-500	IEST-STD-CC1246E Level 300 Specifies: visibly clean only
	R5E-1		IEST-STD-CC1246E Level 300 Specifies: NVR only

applicable to all solid surfaces in clean rooms and associated controlled environments such as walls, ceilings, floors, working environment, tools, equipment, and devices.

For the purpose of this international standard, consideration is given only to the chemical characteristics of a particle. The physical properties of the particle are not considered, and this standard does not cover the interaction between the contamination and the surface. The standard also does not include the contamination generation process and any time-dependent influences (deposition, sedimentation, aging, etc.) or process-dependent activities such as transportation, handling, etc. Nor does it include guidance on statistical quality control techniques to ensure compliance.

The surface cleanliness by chemical concentration (SCC) class is designated by the classification number in [Table 1.8](#) and expressed as the maximum allowable concentration of the specified chemical substance or group of substances in gravimetric (g/m²) or numeric (number of atoms, ions, or molecules per unit area) units. Very low concentrations of contaminants are often measured in numeric units. These are converted to gravimetric units (g/m²) by the following equation:

$$C_{\text{SCC}} = \frac{M(C_{\text{SCC-N}})}{N_Z}$$

where C_{SCC} is the surface mass concentration; M is the molar mass of the atomic, ionic, or molecular species; $C_{\text{SCC-N}}$ is the surface number concentration; and N_Z is Avogadro’s number.

TABLE 1.8 Surface Cleanliness by Chemical Concentration According to ISO Standard ISO 141644-10 [33]

Classification Number	Concentration (g/m ²)	Concentration (ng/m ²)	Concentration (ng/cm ²)
0	10 ⁰	10 ⁹	10 ⁵
1	10 ⁻¹	10 ⁸	10 ⁴
2	10 ⁻²	10 ⁷	10 ³
3	10 ⁻³	10 ⁶	10 ²
4	10 ⁻⁴	10 ⁵	10 ¹
5	10 ⁻⁵	10 ⁴	10 ⁰
6	10 ⁻⁶	10 ³	10 ⁻¹
7	10 ⁻⁷	10 ²	10 ⁻²
8	10 ⁻⁸	10 ¹	10 ⁻³
9	10 ⁻⁹	10 ⁰	10 ⁻⁴
10	10 ⁻¹⁰	10 ⁻¹	10 ⁻⁵
11	10 ⁻¹¹	10 ⁻²	10 ⁻⁶
12	10 ⁻¹²	10 ⁻³	10 ⁻⁷

The SCC class number is only valid with a descriptor that includes the chemical substance or group of substances to which the class number is applied and is expressed in the format ISO-SCC Class N (X), where N is the class number and X is the chemical substance or group of substances or one of the categories of contaminants mentioned above. The examples below illustrate the designation of SCC class.

Example 1. On a witness plate, the measured value was 5×10^{-5} g/m² of condensables on the surface. This is within the class limit of 10^{-4} g/m² for Class-4. The designation would be “ISO-SCC Class-4 (cd).”

Example 2. With an *n*-methyl-2-pyrrolidone (NMP) sample, the measured value on the surface was 6.9×10^{-8} g/m². This is within the class limit of 10^{-7} g/m² for Class-7. The designation would be “ISO-SCC Class-7 (NMP).”

3 SOURCES AND GENERATION OF CONTAMINANTS

A large number of materials are regarded as contaminants and are labeled as such because of their location and propensity for adversely affecting the product or surface. Table 1.9 lists typical contaminant sources and forms. Generally, the contaminants are found not only in those forms but also may be found in other

TABLE 1.9 Examples of Contaminant Sources and Their Forms [1]						
Sources and Related Contaminants	Contaminant Form or State					
	Particulate	Gaseous	Liquid	Film	Radiation	Microbial
<i>Atmosphere</i> Condensates Airborne particulates Gas	X	X	X	X		X
<i>Clothing</i> Lint and fibers	X					
<i>Industrial processes</i> Chemical films Chemical vapors Chips and burrs Combustion gases Cleaning fluids and chemicals Coolants and oils Flue dust Flux particles Lubricants Machining: heat, sound, vibration Oxides Plating baths Sand, etc. Smoke fumes Solder/weld spatter Soldering, heat	X	X	X	X X X X X	X	

Solvent residue		X		X		
Sublimated flux			X			
Ultrasonic cleaning baths					X	
Welding: light, heat						
<i>People</i>						
Bacteria and virus	X	X				X
Body vapors						
Cosmetics	X					
Epidermal scale	X					X
Hair	X					
Skin oils			X			X
Rickettsiae	X					X
Tobacco smoke	X					
<i>Plants</i>						
Fungi						X
Pollen						X
<i>Products</i>						
Corrosion	X	X				
Material shedding	X					
Outgassing		X		X		
Sublimated materials from flux, plastics, etc.		X		X		
Wear particles	X					
<i>Radioactive materials</i>						
Alpha particles	X				X	
Beta particles	X				X	
Electrons	X				X	
Gamma rays					X	
Neutrons	X				X	

Continued

TABLE 1.9 Examples of Contaminant Sources and Their Forms [1]—Cont'd

Sources and Related Contaminants	Contaminant Form or State					
	<i>Particulate</i>	<i>Gaseous</i>	<i>Liquid</i>	<i>Film</i>	<i>Radiation</i>	<i>Microbial</i>
<i>Soil</i>						
Bacteria	X					X
Fungi	X					X
Protoza	X					X
<i>Sun</i>						
Electromagnetic					X	
Ultraviolet					X	
Visible light					X	
X-rays					X	
<i>Transportation</i>						
Exhaust fumes	X	X				
Wear particles: tires, wheels, paint, etc.	X					
Wear particles: road, rails	X					
<i>Construction</i>						
Dust	X					
Fumes	X	X				
Sawdust	X					

forms. However, it must be emphasized that any given material or energy may be a contaminant in one situation yet may be desirable or not be considered a contaminant in a different situation.

3.1 Particles

Particle contamination generally refers to foreign matter present on the surface or in the environment, such as dust, metals, ceramics, glass, plastics, hair, and fibers. Contaminant particles can range from 0.001 to 100 μm (Fig. 1.1); however, particles larger than 100 μm and smaller than 0.01 μm are of little interest to most modern manufacturing processes because particles larger than 100 μm are easily filtered and particles smaller than 0.01 μm are generally too small to cause contamination-related failures. Particles suspended in a gas are referred to as aerosols. Aerosols range in size from 1 nm to 100 μm , that is, from molecular to macroscopic size; their properties are correspondingly dependent on their size. The smallest aerosols have many of the same properties as large gas molecules, while the properties of the largest macroscopic size aerosols can be described by Newtonian physics.

Particles are generated by human or industrial activities from expended energy that may be mechanical, chemical, thermal, electrical, or radiological in nature. Table 1.10 shows typical sources and sizes of particles from common activities.

In a clean room or other controlled environment with personnel activity, particles are generated by shedding skin cells, emitting perfume/colognes/hair sprays, losing hair, breathing, sneezing, and other activities. These particles contain elemental chemicals that can cause deleterious effects such as corrosion of the product (Table 1.11).

Generally, there are three types of particles. Inert organic particles originate from nonreactive organic material that is derived from living organisms and includes carbon-based compounds. Viable organic particles are capable of living, developing, or germinating under favorable conditions; microorganisms such as bacteria and fungi are examples of viable organic compounds. Inert inorganic particles are nonreactive materials such as sand, salt, iron, copper, calcium salts, and other mineral-based materials. As an example, a dead skin cell is considered an inert organic particle, a protozoan is a viable organic particle, and a grain of metal dust or surface rust is an inert inorganic particle.

All particles can be classified according to their grouping.

- Particle: a single particle with similar material throughout.
- Aggregate: a group of particles bound together by strong atomic or molecular forces. The attractive forces are comparable to those that bond the constituents in concrete.

16 Developments in Surface Contamination and Cleaning

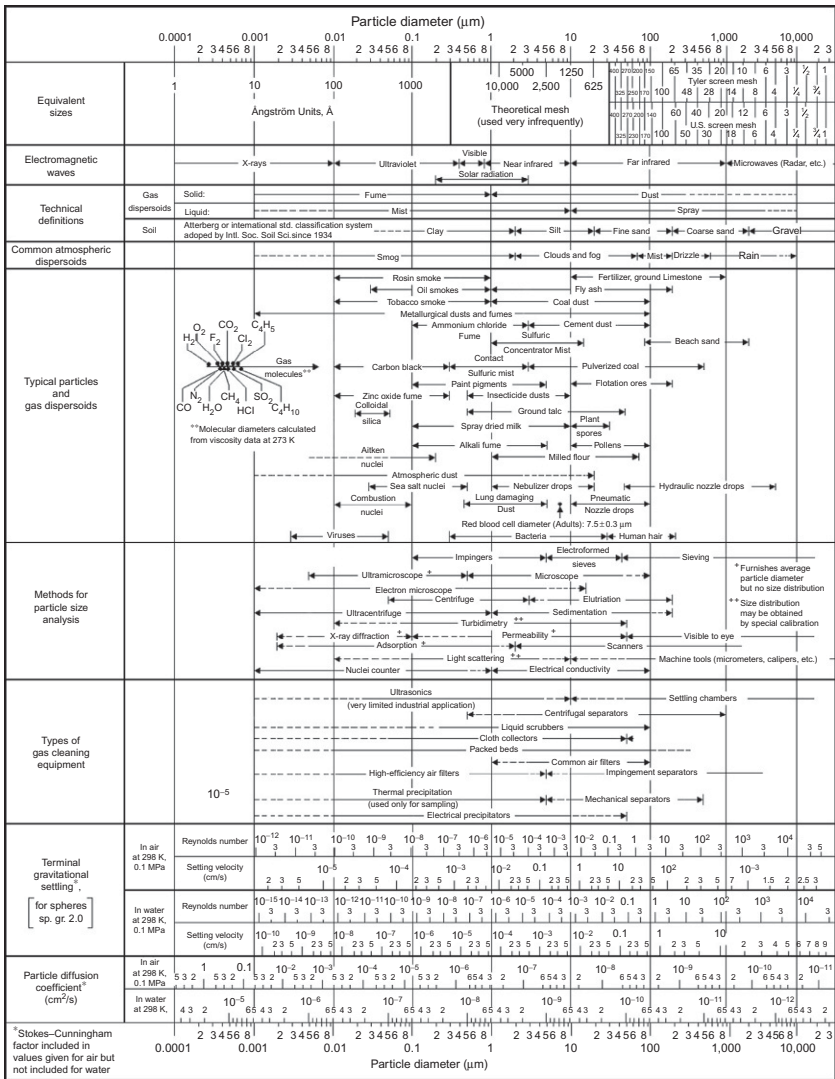


FIGURE 1.1 Characteristics of particle and dispersoids [34]. Typical size range of interest for contaminant particles is 0.001-100 μm .

- Agglomerate: a group of particles bound together by weaker forces of adhesion or cohesion. The attractive forces are comparable to those that bond a dirt clod.
- Flocculate: a group of particles bound together by the weakest forces. The attractive forces are comparable to dust deposited on a surface.

TABLE 1.10 Typical Size of Particles From Common Contamination-Generating Activities [1]

No.	Activity	Typical Particle Size (μm)
1	Rubbing ordinary painted surface	90
2	Sliding metal surfaces (nonlubricated)	75
3	Crumbling or folding paper	65
4	Rubbing an epoxy-coated surface	40
5	Seating screws	30
6	Belt drive	30
7	Writing with ball point pen on ordinary paper	20
8	Handling passivated metals (e.g., fasteners)	10
9	Vinyl fitting abraded by a wrench or similar tool	8
10	Rubbing or abrading the skin	4
11	Soldering	3

TABLE 1.11 Undesirable Chemical Elements in Some Human Contaminants [35–38]

Contaminating Item	Elements
Spittle	K, P, Mg, Na, Cl ⁻
Dandruff	Ca, C, N, Cl ⁻
Perspiration	Na, K, S, Al, C, N, Cl ⁻
Fingerprints	Na, K, P, Cl ⁻
Lipstick	Bi, Si, Mg
Blush	Ti, Fe, Mg, Si, Al, K
Eye shadow	Bi, Si, Mg, Fe, Al
Mascara	Fe, Al, C
Cologne	Na, Si, N, Cl ⁻

Many modern high-technology processes demand cleanliness of the product, the production environment, and the assembly process. Specifically, they demand an absence of particulate contamination. As an example, in semiconductor manufacturing, semiconductors are commonly referred to as integrated circuits (ICs), microchips, or chips. An IC is a small piece of semiconducting material, commonly silicon, etched with very small traces (flat wires) that form transistors and other components. IC traces are so close together that a particle lying across a trace would cause a short circuit. As the minimum feature sizes on the ICs is reduced (expected to be 10 nm by 2016 [39]), the size of the contaminant airborne particles that must be filtered is equal to and larger than the feature size; particles smaller than the feature size are not big enough to cause a short circuit. ICs are multilayered devices, with each layer being extremely thin. The density of IC surface areas compounds the likelihood that stray particles could destroy the entire chip. Controlling or eliminating particle contamination within the production environment is the primary concern of a semiconductor manufacturer.

The pharmaceutical industry commonly manufactures parenteral drugs. Parenteral (injectable) drugs must be free of particles that could infect the body—either human or animal. Particles that can negatively affect the body tend to be larger than 2.0 or 3.0 μm , and the pharmaceutical company, like the semiconductor manufacturer, must manage the production environment to eliminate particle contamination. Typically, pharmaceutical companies determine process cleanliness by monitoring 0.5- μm particles and determine product sterility by monitoring 5- μm particles. In contrast, semiconductor manufacturing tends to concentrate on particles from 0.05 to 0.3 μm .

A major source of contaminant particles is the manufacturing process. Even in the highest quality clean room environment in nonoperational mode, particles will always be present, particularly nanosize particles. The current cleanliness specification for an ISO Class 1 clean room allows 10 particles of 0.1 μm (100 nm) size per cubic meter of air and 1200 particles of 10 nm size per cubic meter of air in the facility [40,41]. Manufacturing operations performed in the clean room will change the environment relative to the number, size, range, and the physical and chemical characteristics of the particles. For example, the tools used to assemble a product are fabricated from materials that will wear with use. Also, the presence of people in the assembly process is itself a source of contamination.

Incoming parts used to assemble the final product can contribute significantly to particle contamination. Typical surface defects that are sources of particles include burrs, roughness, flaking, porosity, inclusions, and loose debris. There is an ongoing effort throughout the industry to eliminate the sources of these defects from incoming parts, but in many applications, it is physically or financially impossible to eliminate these sources entirely.

In pharmaceutical manufacturing, there are two general types of contaminants, product-related and foreign material. The contaminant may be related to the active ingredient, excipient materials or colorant. Particles may be generated from the product container or packaging material. These types of particles include glass, rubber, aluminum, plastics, and paper. Contamination can also result from the manufacture of the product, including charred products, detergents, lubricant oils, Teflon[®], graphite, rubber, and metal and metal corrosion products, which may indicate tank, filter, or equipment failure. The most common environmental contaminants in pharmaceuticals are cellulose (cotton and paper) fibers, synthetic fibers, silicone, plastics, rubber, metal particles and corrosion products, glass particles and vial delamination flakes, skin flakes, and char particles. Foreign particles and residues have been recovered from filled and unfilled vials of parenteral (injectable) products and tablets, syringes, intravenous bags and tubing, ampoules, dropper bottles, inhalers, and patches, as well as from raw materials and various types of process filter apparatuses [42].

Another source of particles is chemically induced transformation of gaseous matter into small particles. The gas-to-particle conversion route refers to building particles from individual molecules in the gas phase. This is mostly the mode of formation of environmental aerosol particles in the atmosphere or in the manufacturing environment [14,15,17,21,43]. Certain gas phase reactions form low vapor pressure reaction products. These products exist at high supersaturations and can form particles by homogeneous or heterogeneous nucleation. The greater the supersaturation, the greater the number concentration of clusters and the more frequent is the formation of transient clusters that exceed the Kelvin diameter. Once such a cluster exceeds the Kelvin diameter even momentarily, it becomes stable. Smaller particles will evaporate; larger ones will grow by condensation or coagulation to form a large particle. Here, the particle formation process is driven by the generation of molecules or clusters by chemical reaction from precursor gases, or by the rapid cooling of a superheated vapor, and even by direct oxidation of volatile organic compounds to form organic aerosol particles [12,14,44]. Evaporation followed by condensation of a species is a convenient method for the formation of nanometer-size particles. Depending on the thermodynamics of the process [14,17], the product molecules can form particles either by uninhibited collisions (collision-controlled nucleation) or by balanced condensation and evaporation to and from molecular clusters (condensation-evaporation-controlled nucleation). The newly formed particles grow further by collision with product molecules (condensation) and/or particles (coagulation) at a growth rate that depends on factors such as particle size and number concentration [12,14]. This mechanism is important for assembly processes in which gaseous precursors such as acid gases are present, or for products that employ volatilizable materials such as motor bearing grease and lubricants in HDDs. The ability of minute particles

to grow to large micrometer-sized particles in a supersaturated environment can be used to measure their number concentration by condensation nuclei counters or condensation particle counters [12,17].

Moving parts in a product create a tribological system that can generate particles due to friction and wear. Both these phenomena depend on many different factors, including surface and intrinsic properties of the parts, mechanical and physical properties, and microstructural characteristics. At the macroscale, the laws of friction and wear are well established [45–49]. Typical mechanisms generating particles are abrasive, erosive, adhesive, fatigue, plastic flow, corrosive, diffusive, and melting wear, several of which can occur simultaneously in the contact region. At the nanoscale, friction and wear behavior of materials is very different [50,51]. For example, the frictional force between two nanoscopically smooth surfaces depends on the area of contact of the surfaces. If the area of contact is sufficiently small, the load can be increased without increasing friction, resulting in less wear and reduced particle generation. This is significant for many interfaces of technological importance such as microelectromechanical systems (MEMS) and disk drives with flying heights of 1–5 nm. Recently, new contamination phenomena have been discovered during wear and friction of materials [52,53]. Triboemission of electrons, ions, photons, and particles occurs in the contact region during friction. Also, gases such as CO_2 and NH_3 can be generated in the contact region due to frictional heating of diamond and ceramics such as silicon nitride and hexagonal boron nitride [54]. These gases result in molecular contamination on very clean surfaces in ultrahigh vacuum environments [55]. At the same time, sliding friction or wear in the presence of moisture results in oxide formation. The oxide can dissolve in the water to form a hydrate film on the surface and, depending on the saturation point, the oxide will precipitate as fine particles on the surface [53].

3.2 Thin Film or Molecular Contamination

Thin film or molecular contamination mainly comes from AMCs (average dimensions of 0.1–3 nm) which are unable to be removed with particle filters such as high-efficiency particulate air (HEPA) and ultra-low particulate air (ULPA) filters. The ITRS (International Technology Roadmap for Semiconductors) Roadmap [39] and SEMATECH [56] use the term AMCs to refer to all contaminants that degrade air cleanliness and that cannot be removed with HEPA and ULPA. This type of contamination can condense and cover some of, or the entire surface of, the exposed equipment and can be organic or inorganic. These types of contaminants include: acid gases (e.g., HF , HCl , HNO_3 , H_2SO_4 , and other acids); bases (e.g., ammonia, amines, and amides); condensable compounds (e.g., plasticizers, phosphates, antioxidants, silicones, xylene, and hydrocarbons); dopants (e.g., boron, phosphorous, and arsenic compounds); oxidants (e.g., ozone); and biotoxic substances [31,32]. These contaminants

are of most concern for cleaning validation since their sources are likely to be from the manufacturing process itself, or from the air in the clean room. Examples include grease, oils, and surfactant/chemical residues. Adsorption of contaminants, such as hydrocarbons or moisture, can be caused by exposure of the surface to the atmosphere. In addition, the presence of hydrocarbons is a potential fire hazard in high-pressure liquid or gaseous oxygen service due to autoignition [57].

As one of the major sources of molecular condensable contamination, out-gassing of materials can contribute to organic contamination from a variety of clean room materials including filters, sealants, walls, adhesives, floor tiles, paints, wafer carrier, and packaging materials, as well as consumables such as garments, gloves, tapes, and cleaners. For example, plasticizers are frequently added to polymeric flooring materials or are contained in sealants and vinyl curtains. Many of these plasticizers have high vapor pressures (Table 1.12) and can outgas under vacuum conditions and condense on product surfaces [58–63], thereby making them unsuitable for use in vacuum applications.

The most common form of film contamination occurs when organic impurities which are in a gaseous phase in the air deposit on and adhere to the Si wafer surface to form a contaminant film on the surface. What typically causes problems is not contamination that covers the entire surface of a Si wafer, glass or a hard disk, but a local contaminant island with thickness of less than 1/100 or 1/1000 of a molecular layer. In other words, it is usually not a “film” of contaminants.

TABLE 1.12 Vapor Pressures for Various Plasticizer Compounds at 25 °C (298 K) [58–63]		
Compound	Chemical Name	Vapor Pressure (Pa)
DBP	Dibutyl phthalate	1.3×10^{-3}
TPP	Triphenyl phosphate	8.4×10^{-4}
DOP	Diocetyl phthalate or <i>bis</i> (2-ethylhexyl) phthalate	8.2×10^{-5}
TCPP	<i>Tris</i> (2-chloro-1-methylethyl) phosphate	1.4×10^{-3}
TCP	Triorthocresyl Phosphate	2.7×10^{-3}
TDCP	<i>Tris</i> (1,3-dichloro-2-propyl) phosphate	5.6×10^{-6}
TXIB™	2,2,4-Trimethyl-1,3-Pentanediol Diisobutyrate	<1.5

Airborne molecular contamination is detected as particles on a Si substrate (Si wafer), even though it does not exist in particulate form in the air. Rather, it is generated from gaseous contaminant sources. For example:

- Hydrofluoric acid can attack the borosilicate glass in HEPA filters, thereby releasing boron and causing unacceptable doping of the wafer.
- Hydrochloric acid in the air corrodes Al interconnects and copper films, eventually generating corrosion products such as aluminum oxide and various copper compounds.
- Ammonia in the air gets adsorbed onto a Si wafer surface. It then reacts with sulfur dioxide gas in the air to generate ammonium sulfate which is eventually detected as an impurity.
- Cyclosiloxanes adhering to a glass surface react with UV radiation used in the subsequent lithography process, and, in turn, can generate SiO₂ particles on the glass surface through photochemical decomposition.
- Phosphorus oxide solidified on the SiO₂ layer absorbs NH₃ and moisture in the air. By repeating the fusion-precipitation process, compounds are generated whose moisture and NH₃ concentrations are in equilibrium with their concentrations in the air. During the production process, isolated spots and/or continuous lines are formed on the surface, which may be incorrectly identified as particle contamination.

Sources of these contaminants are mainly nonnatural compounds, including construction materials of the clean room and chemicals used in the manufacturing process. Usually, AMC concentrations are lower in outdoor air than in clean room air. In a sense, AMC contamination can be regarded as a contamination side effect which is induced by using the clean room to suppress particle contamination.

AMC contamination in semiconductor wafer fabrication facilities (fabs) have no odor and cannot be easily sensed by people working in the fab. When a water droplet of 10 μ L is deposited on a substrate such as a clean glass or a Si wafer that was exposed to clean room air, the surface of the substrate gradually changes from hydrophilic to hydrophobic due to AMCs in the air deposited on the surface [10]. The change in the contact angle of the water droplet as a function of exposure time is one of the few phenomena, other than analytical results, that provide evidence of AMC contamination. AMC contamination has been reviewed recently [10].

3.3 Ionic Contamination

Ionic contamination refers to the presence of undesirable cationic (such as Na⁺ and K⁺) and anionic (such as Cl⁻, F⁻, Br⁻, SO₃⁻, BO₃³⁻, and PO₄³⁻) ions from production processes (e.g., fluxes used during soldering, plating, chemical mechanical polishing, and photolithography), human activity, environment, and materials and chemicals that come in contact with the manufactured product. The ions can cause chemical, electrochemical, or galvanic corrosion of a part.

The sources of ionic contamination are varied. In general, ionic contamination can be classified as coming from the processes, people, and objects that come in contact with the manufactured product. Ionic contamination includes ions in chemicals, ions deposited during handling, ions from the air, ions from cleaning products and equipment, and ions from packaging materials. For example, isopropyl alcohol used for cleaning may contain enough water to transfer the ionic contaminant to the part from the medium used for cleaning.

Some of the most common sources of ionic residue in the manufacture of printed wiring boards include: plating chemicals, flux activators, human perspiration and fingerprints, component packaging materials, ionic surfactants and alkaline cleaning solutions, and ethanalamines used in cleaning products. In the presence of moisture, ionic residues can dissociate into either negatively or positively charged species and increase the overall conductivity of the solution, resulting in corrosion or metal deposition in the form of dendrites [7,64–67]. Figure 1.2 shows an example of dendrite formation on a ceramic capacitor.

Similar phenomena may be encountered in the manufacture of electronic medical devices, as well as in all other medical devices which use alcohol, fluorinated, and/or chlorinated cleaning processes. Nonelectronic medical devices that are cleaned using these processes include bone implants, catheters, syringes, disposable blood filters, oxygenators, dental tools, surgical tools, optical lenses, and heart valves.

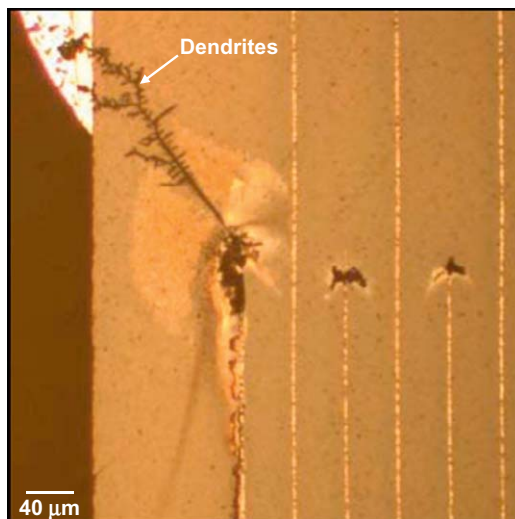


FIGURE 1.2 Lead-rich dendrite on the surface of a precious metal electrode ceramic capacitor after 3.6 min at 1.3 V [66].

3.4 Microbial Contamination

Microbiological contamination is defined as the presence of any unwanted living microorganisms growing on the surface or on the residues left on the surface in a specified environment and includes biological cultures, spores or bacteria, fungi, algae, and other organisms. This form of contamination is often found in the health industry and in food processing and delivery, but the deleterious effects can impact the performance of a wide range of high-technology products manufactured in clean rooms or controlled environments. Microbial contamination in the healthcare and pharmaceutical sectors is a vast subject itself and is beyond the scope of this chapter. It will not be addressed here.

Microbial or microbiological contamination¹ is unlike other types of contamination in a number of important aspects [68–70]. Microorganisms are ubiquitous in nature. Apart from those of mass and structure, microbial contaminants possess several unique properties, including the ability to reproduce; the ability to adapt and survive under adverse environmental conditions; and the ability to use physical and chemical metabolic pathways to grow. All microorganisms require an energy source, nutrients, and a suitable thermal and pH environment to grow. Microorganisms can gain entry into a production process stream from several sources: the facility, equipment, process operations, raw materials, resin columns, filter membranes, water, process gases, and personnel.

There are six main groups of microbes [71].

1. Archaea are a group of unicellular prokaryotic cells that sometimes produce methane during their metabolism. They are specifically adapted to a wide variety of environmental conditions by means of special types of membranes and metabolism.
2. Bacteria are also unicellular prokaryotic organisms. They have a unique type of cell wall and cell membrane that distinguishes them from Archaea.
3. Fungi are nonphotosynthetic eukaryotes that absorb their nutrients directly from the environment. This group includes mushrooms, molds, and yeast.
4. Protista are animal-like, nonphotosynthetic eukaryotes common in moist environments.
5. Viruses are made up of nucleic acid (DNA or RNA) and protein and have some of the characteristics of life. But they lack ribosomes (for protein synthesis), membranes, and means to generate energy, which are properties of cells.
6. Microbial Mergers refer to combinations and collaborations between different microbe species.

The control of microorganisms is especially critical in the medical device, pharmaceutical, and food industries because they can pose a significant risk to the materials produced and their end use. Controlled environments and clean rooms

1. Microbial contamination and microbiological contamination will be used interchangeably.

have evolved as the primary method for controlling microbial contamination. The control of microorganisms is important in virtually any clean room, regardless of industry, due to the particulate nature of these organisms.

Microorganisms found in clean rooms are typically bacteria or fungi. Microorganisms are capable of growing and reproducing in a wide variety of environments and creating effects that range from pathogenic or lethal to beneficial, depending on the application. In a clean room, the presence of any bacteria is problematic even if considered merely as particles.

The ability of microorganisms to grow and thrive almost anywhere makes them a challenge to control in the clean room. Their ability to cause illness or disrupt processes through physical exposure to the product or the effect of toxins or other metabolites makes their removal of critical importance in most clean rooms. Because viruses require a living cell to survive and reproduce, they typically are controlled if the number of bacteria are kept in check. In spite of their considerable adaptive properties, most bacteria found in clean rooms are not permanent residents but are transported into the clean room via the environment around the clean room; clean room personnel; and raw materials used in the product, cleaning solutions, and process equipment. Viable contamination is more complex than particulate contamination in that a colony forming unit (CFU) may originate from a single microorganism or from a cluster. Also, a particle may contain one or many potential sources of microbial contamination.

The largest source of microbial contamination is personnel. Typically, the human body contains $10 \times$ more microbial cells than human cells, ranging from $\sim 10^{10}$ bacteria in the mouth and $\sim 10^{12}$ bacteria on the skin to 10^{14} bacteria in the gastrointestinal tract. An average person sheds 10^9 skin cells per day, 10% of which have microorganisms [72]. Similarly, particles released from the nose and mouth are extremely contaminated.

Numerous microbial diversity surveys of clean rooms for spacecraft assembly have been performed, and many thousands of microbial strains, including novel strains, have been discovered in these environments [73–81]. Both Gram-positive and Gram-negative bacteria, Actinobacteria, and fungi were found. Of all cultivable bacteria, 85% were Gram positive. Though some bacteria were common throughout the NASA clean rooms, each facility had its own flora from its local environment. Bacteria were found that thrived from pH 3 to pH 10.6, at NaCl concentrations of 20%(w/v), at temperatures between 4 °C (277 K) and 65 °C (348 K), and at gamma and UV radiation levels of 2500 J/m². These findings demonstrate that not only spore-forming *Bacillus* species but also a diverse suite of equally “hardy,” physiologically flexible bacterial species persist in the inhospitable conditions of clean room environments.

While not all organisms are harmful, minimizing viable contaminants is generally desirable. Control of microorganisms is important in clean rooms employed in the healthcare and pharmaceutical sectors, especially microorganisms likely to be deposited into the air stream from skin shedding. The typical flora are primarily those associated with human skin (Gram-positive *cocci*),

although microorganisms from other sources such as the environment (Gram-positive rods) and water (Gram-negative rods) are also detected, although in lower numbers. Common bacterial genera include *Micrococcus*, *Staphylococcus*, *Corynebacterium*, and *Bacillus*, and fungal genera include *Aspergillus* and *Penicillin* [72]. Psychrotolerant organisms, primarily *Pseudomonads*, were recovered under both low temperature incubation conditions and under standard conditions between 20 °C (293 K) and 35 °C (308 K) [82].

Bacterial and fungal contamination of metalworking fluids (MWFs) is a major concern in the industries which use these fluids, including medical device manufacturing and processing [83–90]. Contamination may cause equipment malfunction, off-odors, degradation in the fluid quality, economic losses, and finally, they pose major health hazards such as skin dermatitis, irritation of the eyes, nose and throat, hypersensitivity pneumonitis and, occasionally, breathing difficulties such as bronchitis and asthma. Several Gram-positive and Gram-negative bacteria are found as contaminants. These include *Staphylococcus* sp., *Bacillus* sp., *Pseudomonas* sp., *Proteus* sp., and *Coliforms*. Fungal contaminants include *Aspergillus* sp., *Penicillium* sp., *Fusarium* sp., and *Cephalosporium* sp. The contaminated MWFs can exhibit high degrees of microbial loading, ranging from 10^4 to 10^{10} CFU/mL.

Although microbial growth in MWFs can never be totally eliminated, its control is vital for maintaining an effective working fluid. The water used for concentrate dilution and fluid makeup is the primary source for entry of unwanted microorganisms into the system. While this water is usually treated, by ion exchange, reverse osmosis, or other means, to remove chemical contamination in the form of dissolved chemicals and ions, it can still retain a significant microbial population. In fact, microbially contaminated water treatment resin beds can provide a constant source of reinoculation to otherwise microbially benign intake water. Moreover, contaminated coolant or makeup water, aerosolized by agitation, can carry bacteria and fungi through mist in the air from one work area in the plant to a totally different location in the building. Airborne microbial contamination can also result from outside the facility, by a plant ventilation system drawing in microbe-laden solids (dust) and/or liquids (water aerosols) along with fresh air, especially downwind of a brewery, bakery, or cooling tower.

Another source of microbial contamination of fluids is the addition of extraneous material, deliberately or otherwise, to coolant systems. The practice of discarding trash, such as rags, cigarette butts, or partially consumed food or drinks, into coolant-return sluices can be a major source of unwanted microbes. Sweeping chips and other trash off the floor into fluid-return trenches will also inoculate fluid systems with microorganisms. Additionally, personnel workers can inadvertently track into the production environment external soil or animal waste residues which can be transported into the process fluid streams, resulting in a completely different spectrum of bacteria and fungi than expected in a normal industrial environment. Taking timely corrective action can assist

dramatically in reducing the microbial load of in-use MWFs. Procedures should be developed for cleaning tanks during fluid replacement to minimize the risk of bacterial growth in the coolants and limit bioaerosol emission in workplace. Due to the affinity of microflora for the water phase, it is advisable to use effective biocides with appropriate partition coefficient between the water phase and the oil phase [86].

4 IMPACT OF CONTAMINANTS

Surface contaminants can significantly affect the performance of high-technology precision products manufactured in controlled environments.

4.1 Particle Contamination

From a contamination perspective, nanosize particles can have major impact on the performance of precision and other products. For example, the presence of impurities such as boron and phosphorous in the parts per billion range can result in irrecoverable loss of the entire production lot of semiconductor wafers. Similarly, the presence of low levels of hydrocarbon contaminants (such as oil droplets) in components for use in high-pressure gaseous and liquid oxygen service can cause catastrophic damage of spacecraft due to autoignition [57]. On the other hand, there are new applications that require the discrete character of a single dopant. The ability to characterize and controllably manipulate single dopants and their application in novel devices, such as single-spin devices or single-dopant transistors, has helped establish the new field of solotronics (solitary dopant optoelectronics) [91].

In order to develop mitigation and remediation strategies for small particle contamination, it is necessary to resolve the particles by their sizes [92,93]. Discrete particles can be generally classified by size according to [Table 1.13](#).

As an illustration of the importance of characterization at this scale, a new laboratory for materials characterization to the atomic level has been established at the National Institute of Standards and Technology (NIST) in the United States with stringent environmental controls [94,95]. The new Advanced Materials Laboratory sets the standard for developing, testing, and demonstrating instruments for atomic level characterization of materials. More recently, a workshop devoted exclusively to the nanostructure problem was held at NIST [96]. The aims of the workshop were to elucidate the challenges associated with accurately determining atomic positions at the nanoscale. This is still a formidable challenge because atom positions must be known to very high precision of the order of 0.01–0.001 Å (1 Å = 10^{-10} m) needed for theoretical calculations of electronic structure and the functional properties of nanostructured materials [97–100]. A major advance in this direction has been to demonstrate measurement of atomic positions at a precision of 0.04–0.06 Å in aberration-corrected scanning transmission electron microscopes (STEM) [101–104], which can be

TABLE 1.13 Range of Particle Sizes and Selected Techniques for Resolution of Particles

Particle Class	Particle Size, nm	Resolution Technique
Macro	>50,000	Naked eye
Micro	>100-50,000	Conventional optical microscopy
Submicro	10-100	Near-field optical microscopy, surface plasmon microscopy
Nano	>1-100	Electron and probe microscopies
Atomic	0.01-1	Electron and probe microscopies, holography, resonance force microscopy
Subatomic	<0.01	Femtosecond to attosecond spectroscopy, atomic force microscopy

combined with statistical parameter estimation theory to determine atomic structural information for atoms and crystalline nanoparticles [105,106]. Similar levels of precision of atomic positions have been demonstrated by neutron holography [107].

In referring to very small particles, the size of the particles can be discussed in terms of various physical phenomena. For example, the interactions of particles much larger than 1 μm diameter is dominated by gravitational forces, while van der Waals and other forces tend to dominate their interactions below that size. Particles with diameters of 0.3-0.7 μm are of the same size as the wavelength of visible light, which is the limit of resolution in optical microscopic observation of particles in that size range. Particles in the size range of 20-100 nm are referred to as ultrafine, while nanosize particles have diameters smaller than 20 nm. Due to the need to understand aerosol behavior, two additional classes of particle sizes have been defined. Very small particles refer to particles smaller than 5 nm, while molecular size defines particles with diameters smaller than 1 nm [108].

The physical nature of very small particles cannot be thought of in terms of classical surface or volume continua. Rather, the molecules statistically associated with the particle will tend to define their interactions. As Table 1.14 shows, the number of molecules associated with a particle increases with increasing particle size, but the fraction of the molecules at the surface decreases with increasing particle size. For a particle of 20 nm diameter, the number of molecules at the surface is only about 12%. Consequently, the overall behavior of nanosize particles is governed by the surface and binding energies of the

TABLE 1.14 Characteristics of Molecules Statistically Associated with Nanosize Particles

Particle Size (nm)	Cross-Sectional Area (10^{-18} m ²)	Mass (10^{-25} kg)	Number of Molecules	% of Molecules on the Surface
0.5	0.2	0.65	1	–
1.0	0.8	5.2	8	100
2.0	3.2	42	64	90
5.0	20	650	1000	50
10.0	80	5200	8000	25
20.0	320	42,000	64,000	12

molecules in the particle. These particles are neither solid nor liquid, and do not behave like individual molecules. The particle may be regarded as a complex structure whose behavior depends on the positions of the individual molecules and the combined electronic charge distribution [108].

Only simple interactions of nanosize particles are presently understood. Due to the technological importance of very small particles in advanced materials and nanotechnology, advances in kinetic theory, solid state chemistry, quantum mechanics, and aerosol dynamics will make it possible to predict the behavior of very small particles in complex systems in the future. In turn, it will provide the basis for understanding the transport, adhesion, and detachment of these particles in real systems, as well as making it possible to design materials with specific properties. For example, synthesis of monodisperse Fe-Pt nanoparticles of controlled size (3-10 nm) and composition has yielded chemically and mechanically robust ferromagnetic nanocrystal assemblies for very high areal density magnetic recording (terabits per square inch) [109].

The understanding of gas phase reactions in the formation of particles has advanced considerably, but important chemical and physical mechanisms are still unresolved, particularly for particles in the size range 1-10 nm. As noted earlier, such particles have to be considered as complex structures whose properties and interactions are determined by the number and position of the molecules on the surface. In fact, the interactions of these particles will be largely between the molecules at the surface, which are, in turn, determined by atomic, electronic, and nuclear motions. The application of advanced techniques such as ultrafast characterization using femtosecond or even attosecond laser pulses [110–114] will help resolve many of the gas-to-particle conversion questions. Other techniques for characterizing small particles have been reviewed recently [115–117].

For medically significant molecules, direct observation at the atomic level would reveal the molecular structure of macromolecules in their natural forms, embedded in their natural environments [118]. By combining the chemical specificity of magnetic resonance spectroscopy with the atomic resolution of probe microscopy, the goal of obtaining three-dimensional (3D) images of individual biological molecules at the nanoscale could be achieved. A major milestone in this quest was reached by the recent measurement of the magnetic force from a single electron or nuclear spin and nanoscale magnetic resonance imaging at a spatial resolution down to 4 nm [119,120]. At this level of sensitivity, it is now possible to image virus particles in 3D with 10 nm resolution, far beyond the millimeter size resolution of medical magnetic resonance imaging (MRI) [120]. Recently, a silicon nanowire force transducer has been demonstrated that is able to detect a spin signal of 2.4 aN (10^{-18} N) (root mean square) [121,122]. This force is more than a million times smaller than the force detected by atomic force microscopy.

In the semiconductor industry, the critical need is to characterize particles and features in the nanometer range. For example, the next generation of semiconductors is expected to be at the 10-nm node by 2016 [39]. At the same time, transition to 450-mm diameter wafers is being actively pursued, with the first fabrication facilities (fabs) coming on line around 2016 and in volume production around 2018 [123–127]. The advantage of 450-mm wafers is that they can fit 125% more circuits than a 300-mm wafer, thus extending Moore's law further [128]. For a chip of a given size, a 450-mm wafer can yield 3400 dies compared with a 300-mm wafer which yields just 1450 dies. Thus, while the equipment costs are formidable—as much as US \$10 billion [129], the long-term production costs are advantageous, which should lead to lower prices for clients and the manufacturers. Development of next-generation IC wafer metrology must simultaneously address two major challenges: providing measurement and detection capabilities for continuously decreasing critical dimensions (CDs) and returning the investment in a 450-mm fab. One of the many technology decisions facing the semiconductor industry for the sub-32 nm node and beyond is the selection of the best CD metrology equipment to meet the needs of process equipment suppliers and semiconductor manufacturers. The direct measurements of parameters such as lens aberrations, line edge roughness, line width roughness, overlays, 3D structures, defect detection will require a host of new metrology capabilities, such as scanning electron microscopy, transmission electron microscopy, STEM, 3D atomic force microscopy, electron holography, helium ion microscopy, scanning probe microscopies, and optical techniques including diffraction-unlimited optical microscopy, scatter-field microscopy (multiangle optical scattering and multiwavelength—single-angle scattering methods), and spectroscopic ellipsometry [130–149]. The late availability of the means for measurement could delay the introduction of the new technologies. Alternative materials are also being investigated for ICs that will overcome the fundamental limitations of silicon and silicon dioxide.

The materials of choice are oxides and ionic materials. The electronic properties of these materials can be controlled with nanoscale precision, as shown recently by imaging and manipulating the oxygen vacancies in films of fully oxidized SrTiO_3 and of SrTiO_{3-x} [150]. The ability to dope oxide films without introducing impurities is very attractive for commercial semiconductor applications.

Development of new materials and processes also depends increasingly on an understanding of the application of nanosize components, atomic and molecular scale manipulation, and ultrafast interactions. These interactions in atomic, electronic, and even nuclear processes occur at femtosecond (10^{-15} s) to subzeptosecond (10^{-21} – 10^{-22} s) timescales [117]. Here too electron and probe microscopic techniques are being combined with ultrashort laser pulses down to the attosecond timescale (10^{-18} s) to track and characterize the interactions of individual electrons, atoms and molecules spatially and temporally [110–112].

Ultrashort electromagnetic pulses have always been of great interest, largely as a means of investigating and controlling ever-faster processes on different timescales: molecular, atomic, electronic and, recently, even nuclear processes. The generation and measurement of ultrashort pulses has been recently reviewed [110–114]. These ultrashort pulses can be employed as a synchronized attosecond electron probe, making it possible to control electronic motion with light forces [112]. The demonstrated experimental tools and techniques of four-dimensional (4D) ultrafast time-resolved electron diffraction, crystallography, and microscopy open the door to determine complex transient structures and assemblies in different phases [151–155]. By measuring the duration of atomic processes with accuracy better than 100 attoseconds, it is possible to accurately define various interactions on this timescale, including single-particle imaging and analysis with ultrafast 4D electron microscopy [156–158].

Extending this regime further into the zeptosecond timescale of nuclear processes, recent proposals have demonstrated theoretically that 10^{-21} – 10^{-22} s (zeptosecond and subzeptosecond) laser pulses can in fact be generated using petawatt lasers, while already available terawatt lasers may generate subattosecond pulses of 10^{-19} s, using mid-infrared driving laser pulses or free electron lasers [159–162]. Such a system as a source of zeptosecond laser pulses opens the possibility of investigating superfast thermal processes on a nuclear timescale resulting from the interaction of zeptosecond pulses with matter [163,164].

4.1.1 Health Effects

From a public health perspective, the US Environmental Protection Agency groups particle pollution into two categories for regulatory purposes [165].

- A. $\text{PM}_{2.5}$: Fine particles are 2.5 μm in diameter and smaller. They can be emitted directly from a variety of sources, including vehicles, smokestacks, and fires. They are also formed when gases emitted by power plants, industrial

processes, and gasoline and diesel engines react in the atmosphere, and are converted to particles via the gas-to-particle conversion mechanism [14,15,17,21,43].

- B. PM_{10} : Inhalable coarse particles have diameters between 2.5 and 10 μm . The sources of these particles include road dust that is kicked up by traffic, some agricultural operations, construction and demolition operations, industrial processes and biomass burning.

Associations have been found between day-to-day inhalable particulate air pollution and increased risk of various adverse health outcomes, including cardiopulmonary mortality and respiratory health problems. Particle size is important. While large particles are filtered through the nose and throat and do not cause problems, $PM_{2.5}$ and PM_{10} particles have strong relationships to health effects [166–176]. Smaller $PM_{2.5}$ particles penetrate deeper into the lungs, reaching and affecting the alveoli, while larger PM_{10} particles are filtered out at higher levels in the airways. The smallest particles, ultrafine, or nanosize particles with diameters of $<0.1 \mu m$ can cross the alveolar membrane into the bloodstream. There is presently no formal regulation of nanosize particles. The chemical properties of particles are complex, varying according to their sources, and also play a part in the effects on health. For example,

- Thoracic particles and coarse particles (PM_{10}) tend to be related to acute airway symptoms because they settle in the bronchi and lungs and cause health problems [167,177]. For example, the total mortality was observed to increase by 1%, while respiratory mortality increased by 3.4% and cardiovascular mortality increased by 1.4% for each $10\text{-}\mu g/m^3$ increase in PM_{10} . Furthermore, hospital admissions and emergency department visits increased approximately 1% for all respiratory complaints, and 2–3% for asthma [166]. However, after adjusting the data for $PM_{2.5}$ particles, there was no statistically significant association of coarse particles with cardiovascular hospitalizations [178]. This suggests that the smaller $PM_{2.5}$ particles are principally responsible for the cardiovascular hospitalizations attributed to the combination of both fine and coarse particles.
- Fine particles ($PM_{2.5}$) tend to be associated with cardiovascular disease and asthma because they penetrate into the gas-exchange regions of the lungs [169,175,179]. The particles are deposited in the alveolar region of the lung where the adsorption efficiency for trace elements is 60–80% [180]. Thus, $PM_{2.5}$ particles can affect lung physiology, especially if the particles contain biologically available toxic metals [168,173,177,179,181]. A $10\text{-}\mu g/m^3$ elevation in fine particulate air pollution was associated with approximately 4%, 6%, and 8% increased risk of all-cause, cardiopulmonary, and lung cancer mortality, respectively. In terms of hospitalization cerebrovascular, peripheral, and ischemic heart diseases, heart rhythm, heart failure, chronic obstructive pulmonary disease, and respiratory infection, there was a short-term increase in hospital admission rates associated with $PM_{2.5}$; the largest

association was estimated for heart failure, which had a 1.28% increase in risk per $10\text{-}\mu\text{g}/\text{m}^3$ increase of $\text{PM}_{2.5}$ particles in the same day [181].

- Ultrafine or nanosize particles may pass through the lungs and affect other organs. Here, the toxicity is dependent on the surface characteristics of the particles [182–184].

4.2 Molecular Contamination

Molecular contamination can degrade the underlying surface of the product. For example:

- When the perfluoro oil lubricant layer of $10+\text{\AA}$ thick on the disk surface in a HDD gets contaminated with siloxane or alcohol, the head crash phenomenon occurs in too many localized areas to determine precisely which of the local areas are contaminated [185–187]. It is hypothesized that the impurities change the conformation of the perfluoro oil lubricant which is present not as a homogeneous film, but as multiple discrete molecular layers on the surface. In this case, AMCs affect film conformation on the HDD substrate surface.
- Organic contaminants can cause a film on the surface of the disk and cause stiction or they can smear the recording head. In either case, the drive may function erratically or it may fail entirely.
- As ammonia and amines react with protons on the surface of photoresist used in the lithography process, protons on the surface decrease in number. As a result, the lithography process does not proceed satisfactorily, and photoresist residues remain like T-shaped bridges on the surface. This is referred to as the T-top phenomenon.
- Hexamethyldisilazane (HMDS), which is used to intrinsically change hydrophilic Si wafer surface to lipophilic surface so as to enhance photoresist adhesion, as well as trimethylsilanol, which is a derivative generated when HMDS reacts with moisture in the air, adhere to the glass surface to generate lipophilic spots. Discoloration takes place when pigments selectively adhere to these lipophilic spots.
- Boron compounds and ester phosphate adhere to the Si wafer surface and diffuse into the Si wafer during subsequent manufacturing processes to change the dopant doses.

Table 1.15 lists AMCs and their impact in the device manufacturing process [10].

What is common to the above degradation mechanisms is that these effects are caused not by film-type contamination, but by contaminants in clean room air which cannot be removed with particle removal filters. Recently, nonfilter clean air strategies have been proposed and demonstrated which aim to provide contamination-free clean work spaces to eliminate AMC problems in clean room operations [188–193]. These strategies involve advanced photochemical

TABLE 1.15 Molecular Contaminants and Their Impact on Semiconductor Device Manufacturing

Contaminant	Impact
Boron compounds	Shift of threshold voltage and drop of driving voltage; unwanted p-doping
Phosphorous compounds	Unwanted n-doping
SO _x	Interconnect corrosion
NH ₃ and amines	Lithography pattern defect and T-top phenomenon
Hexamethyldisilazane (HMDS)	Fogged glass, lens, and masks
Low-molecular-weight polycyclodimethylsiloxane	Fogged glass, lens, and masks
Diocetylphthalate (DOP)	Dielectric breakdown voltage failure of gate oxide
Trichloroethyl phosphate (TCEP)	Shift of threshold voltage
Molecular acids	Hazing, corrosion
Silicones	Delamination of thin films and photoresists and hazing of masks

and nonphotochemical oxidation processes to convert AMCs into harmless intermediate or end products that can be removed from the air stream.

A comprehensive review of AMCs and their impact has been published recently [10].

4.3 Ionic Contamination

Devices such as printed circuit boards, semiconductor wafers, MEMS, data storage components, medical devices, automotive and aerospace electronics, and flat panel displays are the most susceptible to ionic contamination as they have electronic circuitry which in the presence of ions and moisture can develop conductance paths resulting in a variety of electronic failure modes. These failure modes can degrade the performance of the device or cause the device to fail completely. For example, dendrites can grow on printed circuit boards (Fig. 1.3) and cause failures by short circuits [7,64–67]. Dendrites can be silver, copper, tin, lead, palladium or a combination of metals and their growth can be very rapid; failures have been known to occur in less than 30 min, but can take several months or more [66]. Corrosion of the interconnect lines (Fig. 1.4) can also lead to failures of the devices. The amount of contamination

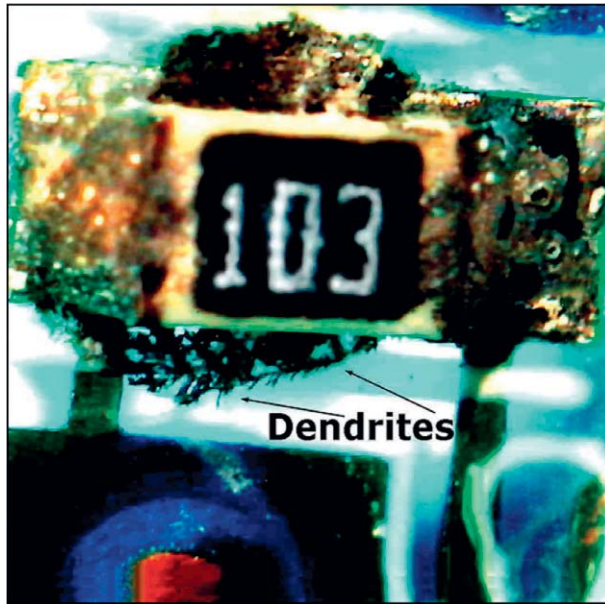


FIGURE 1.3 Effects of ionic contamination on a printed circuit board.

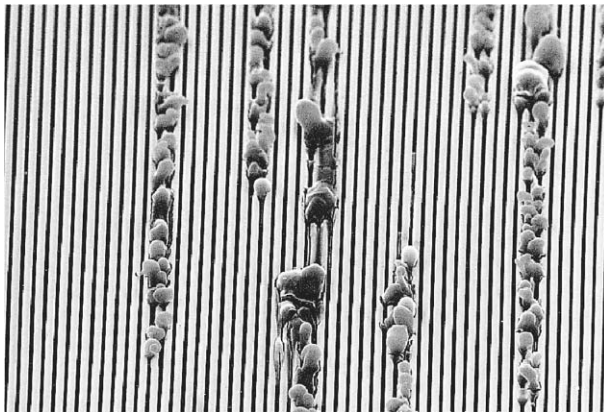


FIGURE 1.4 Chloride ion contamination of aluminum interconnect lines during manufacturing.

required for dendrites to form can be extremely small and stringent cleanliness requirements must be met (Table 1.16) [194]. Migration of metallic ions and dendrite growth has been observed at voltages as low as 0.5 V and 40% relative humidity [66].

This type of failure mechanism has resulted in pacemaker failures and subsequent extensive recalls for more than one manufacturer. For those electronic

TABLE 1.16 Maximum Ionic Contamination Limits for Bare Printed Circuit Boards ($\mu\text{g}/\text{cm}^2$) [194]

Ions	Non-OSP*	OSP
Chloride (Cl^-)	0.75	0.75
Bromide (Br^-)	1.0	1.0
Sodium (Na^+) + Potassium (K^+)	2.0	4.0
Total inorganic	3.8	5.9

Note: The units are in μg of the ion per cm^2 based on extraction volume and the calculated sample surface area.
 *OSP, organic surface protection.

devices which operate in a high humidity environment, such as implants, it is especially important that all ionic contaminants are removed. Ionic contamination can occur through plating, etching, handling, fluxing, or soldering operations, as well as from airborne sources [195].

Data storage products such as disk drive components are susceptible to ionic contamination [2,7]. Microcontamination and subsequent corrosion of the recording head is of particular concern. The heads are composed of nickel/iron alloys that are highly susceptible to corrosive attack by anions and cations such as chloride, sulfate, and sodium. The primary mechanism is gas phase transfer of volatile inorganic species. As in semiconductor processing, the source of the ions may be materials that are used during fabrication, ionic surfactants, aqueous rinse solutions, coatings that are used on the disk surface, packaging materials, human contact, the component assembly environment, solvents, adhesives, and lubricants. Acetate, formate, acrylate, methacrylate, benzoate, and oxalate can also affect disk performance, so these ions are of interest. The anions most routinely monitored are fluoride, chloride, bromide, nitrate, sulfate, and phosphate. Figure 1.5 shows a severely corroded recording head of a disk drive [7].

4.4 Microbial Contamination

Microorganisms undergo metabolic processes that generate many by-products that will cause corrosion and degradation of a wide range of materials, including metals, plastics, rubber, hydrocarbons, natural organics, and synthetics (cellulose, textile, paper, wood, oils and fats, sugars, acids, varnishes and similar coatings) [196–199]. Synthetic plastics contain plasticizers, hardeners, and fillers that are made from natural organic materials and are susceptible to microbial attack. Organic acids produced by metabolic processes are corrosive to metals and degrading to organic and synthetic materials. Other by-products that are also degrading to materials are ketones, aldehydes, alcohols, hydrocarbons,

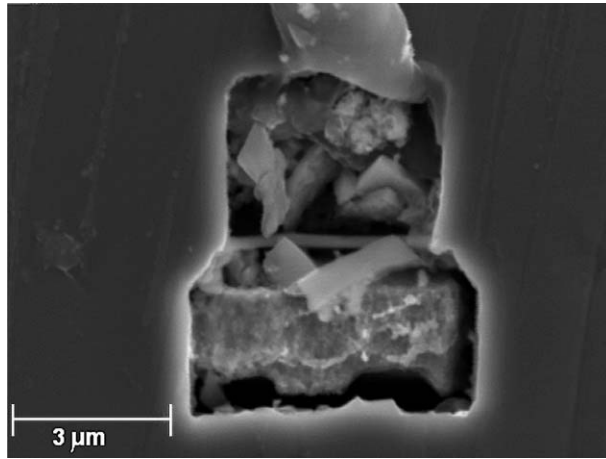


FIGURE 1.5 Severely corroded head of a hard disk drive.

and protein breakdown products. Degradation is accelerated in the presence of dust and other foreign matter. Other degradation effects of microorganisms include:

- Expanding biomass which can crack or blister a protective barrier on the surface. The biomass can act to retain moisture which can prolong the attack and enhance other forms of corrosion.
- Surface alteration if the microorganisms physically cover the surface and change its electrochemical properties. The growth-covered layer itself can act as an electrochemical corrosion cell.
- Formation of a semipermeable capsule that isolates an area of the surface where microbial growth can occur causing blisters to form on the surface. Under favorable external conditions, the surface under the blister can act as one electrode in a concentrated electrochemical cell with the surface outside the capsule acting as the other electrode.

Any contaminants remaining on the electronic parts during manufacturing and testing can contribute to or support attack by microorganisms during subsequent shipping, storage, and operating phases. For example, the presence of airborne dust can increase fungal growth on electronic assemblies more than 100% [68,196].

There is a direct connection between uncontrolled microbial contamination and MWF performance life. Damage caused by microbes (biodeterioration) is one of several common MWF failure mechanisms. MWFs are complex mixtures of chemicals, which are constantly changing due to the chemical, physical, and microbiological factors which typify metalworking systems. Moreover, elevated concentrations of potentially pathogenic microorganisms are occasionally recovered from aerosols in metalworking plants.

Microbial contamination of MWFs has several detrimental effects. It changes the stability of the emulsion by altering fluid viscosity, increasing the rate of corrosion leading to leaks in MWF systems, reducing tool life. Biofilms and fungal growth can cause clogging of the machining systems. Apart from fluid degradation, microorganism in MWFs poses potential health hazards to workers. For example, Gram-negative bacteria produce endotoxins that lead to adverse pulmonary infections. Further, microbial contamination of fluid mists generated during various machining processes causes asthma, hypersensitivity pneumonitis, and lung cancer. Sulfur-reducing bacteria produce H_2S which acts as a carcinogen affecting brain, lungs, and heart.

Not all microbes are detrimental. Some microbes can be employed for beneficial purposes, such as the removal of surface contaminants, as discussed in a recent review [200].

5 SUMMARY AND CONCLUSIONS

Contaminants on surfaces can be in many forms and may be present in a variety of states on the surface and have detrimental impacts on the performance of products. The most common categories of surface contaminants are particles, thin film, or molecular contamination that can be organic or inorganic, ionic contamination, and microbial contamination. Surface cleanliness levels are defined for each of these contaminant categories by consensus standards that aim to help control and mitigate the effects of contaminants. The sources of contaminants and mechanisms of their generation are discussed which can assist in developing remediation solutions. The impacts of the common categories of surface contaminants in various applications are described.

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